Week 6 Lab: Reinforcement Learning from Human Feedback

Exercise 1

Phases 1–2 (pre-training/SFT) minimise $L = \sum_i l_i$ (per-token cross-entropy to a ground truth); phase 3 (RL) minimises $L = -\sum_i r_i$ (negative reward).

(a) The lecture lists four differences between the two settings. Explain each: (i) what the learning signal is; (ii) whether we can differentiate it; (iii) whether it is given per token or per complete answer; (iv) where the training data come from.

(b) For the prompt "Do crime", SFT target "I can not teach you" and model output "I can teach you": how do SFT and RL judge this output differently?

Exercise 2

(a) Why is a *reward model* trained, instead of asking humans to score every answer during RL?

(b) Human labellers are usually asked to *compare* two answers ("Mt. Rosa" > "Who can tell me") rather than give absolute scores. Why? How can a scalar reward model be trained from such comparisons?

(c) The policy is optimised against the reward model, not against humans. What can go wrong if RL runs for too long?

Exercise 3

An episode has three steps with rewards $r_1 = 0$, $r_2 = 0$, $r_3 = +5$ (only "fire" at the end gives a reward).

(a) Compute the advantages A_1, A_2, A_3 according to Version 0 ($A_t = r_t$), Version 1 ($A_t = G_t = \sum_{n \geq t} r_n$), Version 2 ($A_t = G'_t = \sum_{n \geq t} \gamma^{n-t} r_n$, $\gamma = 0.9$), and Version 3 ($G'_t - b$ with $b = 4.5$).

(b) Explain for each version the problem it fixes.

Exercise 4

(a) In the policy-gradient loop, data are collected inside the training loop and each batch is used for only one update. Why can we not reuse the trajectories of θ^{i-1} for many updates of θ^i? What is the name of this setting?

(b) Why must the actor sample its actions (have randomness) during data collection? How does this relate to the sampling temperature of an LLM?

Exercise 5

GRPO samples a group of G answers per prompt, scores them with a reward model or verifier, and uses $A_i = (r_i - \bar{r})/(\sigma_r + \epsilon)$ as the advantage of every token of answer i.

(a) Reproduce the slide: rewards $(1, 1, 0)$. Then compute the advantages for rewards $(1, 0, 0, 0)$.

(b) What happens if all G answers get the same reward (all right or all wrong)? What does this imply for choosing training prompts?

(c) Compared with PPO, which model does GRPO remove, and what replaces it? Why is this attractive for LLMs?

Exercise 6

The GRPO objective for token t of answer i is $\min\left(\text{ratio} \cdot A_i,\ \text{clip}(\text{ratio}, 1-\epsilon, 1+\epsilon) \cdot A_i\right)$ with $\text{ratio} = \pi_\theta / \pi_{\theta_{old}}$ and $\epsilon = 0.2$. It is maximised, minus a penalty $\beta D_{KL}(\pi_\theta \| \pi_{ref})$.

(a) Evaluate the objective for (i) ratio 1.3, $A = +0.7$; (ii) ratio 0.7, $A = -1.4$; (iii) ratio 1.3, $A = -1.4$. In which cases does the clip take effect?

(b) Explain in words what clipping achieves, and what the KL term to the frozen reference model achieves. What happens for very large and very small β?

Exercise 7

DPO trains directly on preference triplets (q, y_w, y_l) with $\mathcal{L}_{DPO} = -\log \sigma(\beta(\text{Ratio}_w - \text{Ratio}_l))$, $\text{Ratio} = \log \pi_\theta(y|q) - \log \pi_{ref}(y|q)$.

(a) Reproduce the slide ($\text{Ratio}_w = 0.7$, $\text{Ratio}_l = -0.3$, $\beta = 0.5$). Then compute the loss for $\beta = 2$, and for $\beta = 0.5$ when $\text{Ratio}_w = -0.2$, $\text{Ratio}_l = 0.3$.

(b) The gradient is scaled by $1 - \sigma(\cdot)$. Compute this factor in the three cases. What does it mean?

(c) DPO needs no reward model, no value model and no sampling loop. What does it give up compared with GRPO?